Aurick Anwar

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Education

McMaster University Hamilton, Ontario
Bachelor of Engineering: Engineering Physics *Expected 2029*

Work Experience

Software Engineering Intern Toronto, Ontario
HermesAI *May 2026 - Present*

- Developed **React.js** Front-end features for an AI-legal tech platform, improving lawyer workflow efficiency by **35%** across **1000+** page PDF case files.
- Built scalable legal document dashboards that reduced manual data processing time by ~**50%** for Canadian law firms.
- Shipped Front-end features supporting **100+ users** across high volume and improved platform responsiveness across high volume legal workflows.

Founding Engineer Toronto, Ontario
Magnified Systems *February 2026 -Present*

- Developing a **mounted device** that can be attached to a helmet and is used to detect and quantify **impact severity** in real time.
- Built a **PyTorch** regression model that predicted an impact severity score from 1-100 at a **96.4%** accuracy.
- Processed a **6-axis IMU** sensor and, using **C++**, developed a severity score algorithm based off linear and angular acceleration of the impact.

Assistant Instructor Toronto, Ontario
The STEAM Project June 2025 – August 2025

- Co-developed a Battle Bots curriculum, teaching **SolidWorks** and **Arduino** to **50+ students** improving project completion rates by ~**40%.**
- Assisted in packaging and delivering **100+ project kits** across multiple locations, ensuring timely distribution.

Media & Operations Lead Toronto, Ontario
Orbitview December 2023 – June 2025

- Led **15+ developers** to build scalable media platforms and AI webinars with **200+ attendees** and **5× ROI**.
- Directed AI-focused media and tech events, generating **$1,500+** in revenue through startup collaborations.

Personal Projects

Autonomous Self-Driving Vehicle with CARLA, [*CARLA, YOLOv11, PyTorch, OpenCV]*

- Developed an end-to-end autonomous driving pipeline in the **CARLA** simulator integrating **YOLO-based object detection** and **vehicle control,** achieving a real-time simulation at ~**20-30 FPS**.
- Wrote to a dataset with **5+ features** (steer, brake, speed, acceleration, throttle), generating **25k+** sample datasets to train a **PyTorch** model.
- Used **Cross Entropy Loss**, achieving ~**88.5%** prediction directional accuracy for precise automation navigation.

AI Smart Computer w/ Hand Gesture Control, [*MediaPipe, PyAutoGUI, OpenCV]*

- Implemented **6+ gesture mappings** with 21-point hand tracking, enabling click, cursor, mute, and volume control via distance thresholds.
- Reduced unintended inputs by ~**60%** using gesture filtering and cooldown logic, achieving **<100 ms** latency.
- Tested **10+ users** with dexterity issues, improving task speed by ~**45%,** while reducing mis-gesture errors by **70%.**

Google Home Replica, [*Python, Google Cloud, Text-to-speech]*

- Built a Google Home/Alexa replica using **Google Cloud APIs** and **Text-to-speech**, processing user queries with **<3s** response time.
- Implemented **4+ command types** (math calculations, music, weather), enabling real time voice interaction.

Technical Skills

Programming Skills: Python, C++, React.JS, CSS, HTML, JavaScript, Git
Libraries: PyTorch, OpenCV, MediaPipe, PyAutoGUI, YOLOv11
3D Design: AutoCAD, Fusion360, Inventor, SolidWorks
Electronics and Microcontrollers: Arduino, Raspberry PI, ESP32